

RNA-X: Modeling RNA interactions to design binder RNA and simultaneously target multiple molecules of different types

Sobhan Shukueian Tabrizi, Helyasadat Hashemi Aghdam, and A. Ercument Cicek[‡]

Department of Computer Engineering, Bilkent University, Ankara, Türkiye

[‡] correspondence: cicek@cs.bilkent.edu.tr

Abstract. RNA interactions with proteins, other RNA molecules, and DNA play essential roles in numerous cellular processes and underpin a wide range of therapeutic mechanisms. Consequently, modeling these interactions is critical for understanding biological systems and designing novel RNA-based therapeutics. However, designing RNA sequences that selectively and strongly bind to a specific target remains a major challenge due to the vast sequence space and the current limitations of experimental and computational methods. In this study, we introduce RNA-X, the first RNA interaction foundation model which is based on masked language modeling for representation learning, conditional RNA design, and optimization. For the first time, RNA-X enables (i) RNA design targeting not only proteins but also other RNA and DNA molecules, and (ii) simultaneous design against multiple targets. Our extensive experiments demonstrate that RNA-X generates RNA sequences with natural structural characteristics and surpasses state-of-the-art methods in protein targeting. Using affinity predictors and molecular dynamics simulations, we further show that the model can design RNA molecules that (i) target therapeutically relevant molecules such as p53 and thrombin proteins, and (ii) bind to targets with no prior interaction data, such as SDAD1 protein. As a proof of concept, we designed a novel guide RNA from scratch that simultaneously binds (i) to the DNA of a bacterium and (ii) to the Cas9 protein. The resulting design achieves a predicted binding energy comparable to that of the wild-type guide RNA reported in the Protein Data Bank. Despite having orders of magnitude fewer parameters than existing RNA foundation models, RNA-X produces representations that outperform them across diverse RNA interaction-related downstream tasks. The code and pretrained model are publicly available on GitHub.

1 Introduction

RNA interactions play a central role in various biological systems such as (i) microRNA–mRNA interactions for post-transcriptional regulation [5], (ii) long noncoding RNA (lncRNA) - DNA interactions for the formation of molecular scaffolds to change chromatin structure [33], and (iii) RNA - protein interactions to regulate splicing [40] and transport [41] [15]. Consequently, the design of novel RNA sequences that can interact with target molecules presents ample opportunities in therapeutic applications.

The therapeutic potential of RNA design through binding to a target X (hence the name RNA-X) has many examples for various molecule types. **RNA-RNA.** In RNA interference (RNAi), small interfering RNAs (siRNAs) silence disease-related genes through degradation of their precursor mRNA. Yet, the design of potent siRNAs remains challenging due to requirements for high efficacy, low off-target effects, and minimal immune activation [1]. **RNA-DNA.** CRISPR–Cas systems rely on single-guide RNAs (sgRNAs) to direct Cas proteins to genomic loci. Ensuring sgRNA efficacy and specificity in design is crucial and challenging to ensure reliable gene editing. **RNA-protein.** RNA binding proteins (RBPs) offer a strategy to inhibit or enhance RNA–protein interactions. This approach expands the current drug landscape, as only a small fraction of human proteins are targetable using ligand-based drugs[19]. Despite its potential, designing RNA molecules that selectively bind to different types of targets remains an open challenge, mostly because of the vast search space and stringent efficacy and specificity requirements. Current practice in the lab relies heavily on trial and error such as high-throughput screening techniques such as SELEX (Systematic Evolution of Ligands by EXponential Enrichment) [6]. While computational methods can help designing RNA to bind with other molecules process, they have critical limitations.

Recent computational research on RNA is focused on the building of RNA foundation models [7,31,46,50]. However, these models are not generative and cannot directly design new sequences. Early generative methods focused on inverse folding, where sequences are designed to fold into a desired secondary structure [18,14,2,17,36,11,9]. Various new deep learning-based models have been proposed to design the RNA structure or sequence that resembles natural RNA [13,34,39,23,4,20,45]. However, these methods cannot condition the design to bind with a target. RNAFlow is a diffusion-based model that can design RNA structure conditioned on a target protein [29]. Yet, due to lack of ground-truth 3D RNA structure data, its designs lack novelty which is a common problem of 3D structure designers. RNAGen [30] and GenerRNA [49] are earlier examples of RNA sequence designers to target a protein which condition for the target only in post-processing, using affinity predictors to guide generation or fine-tuning the model for the target using interaction data. This creates key limitations: (i) the generation and optimization are decoupled, preventing joint RNA–target representations; (ii) target-specific predictors have a narrow coverage and for many proteins there are no known RNA-interactions: For example, DeepCLIP [16] supports only 221 proteins; and (iii) the design of RNAs for new or synthetic targets is infeasible without existing interaction data. Recent approaches focus on directly modeling RNA-protein interactions to avoid these problems. BAnG uses protein structure and sequence as input and trains a model on 3D RNA-protein complex structure data in the Protein Data-bank (PDB) [25]. RNAttranslator is a language model that frames the conditional RNA design problem as a sequence-to-sequence natural language translation problem [37]. Despite progress, all of the methods mentioned above can design RNA to bind to a single target type (proteins) and their embeddings cannot be used for downstream RNA interaction-related tasks, as they are not designed as foundation models for representation learning.

In this study, we present RNA-X, the first RNA interaction foundation model. RNA-X can design an RNA sequence to target RNA, DNA, and proteins; and can design a single RNA molecule to simultaneously bind to multiple targets for the first time. RNA-X, is a masked language model trained end-to-end on more than hundred million RNA-target interactions. It learns joint representations of RNA and target molecules, enabling direct sampling of RNA sequences with desired binding properties. We formulate the problem of target-conditional RNA design as a context-aware generative task based on a target-specific embedding. RNA sequences are generated using techniques like iterative unmasking and refinement of the tokens coupled with a Monte Carlo tree search informed by model’s rewards. The model representations are further optimized to handle the representation degeneration in sequence level representations using whitening and uniformity for

increased performance in downstream tasks without changing model weights. See Figure 1 for RNA-X system model.

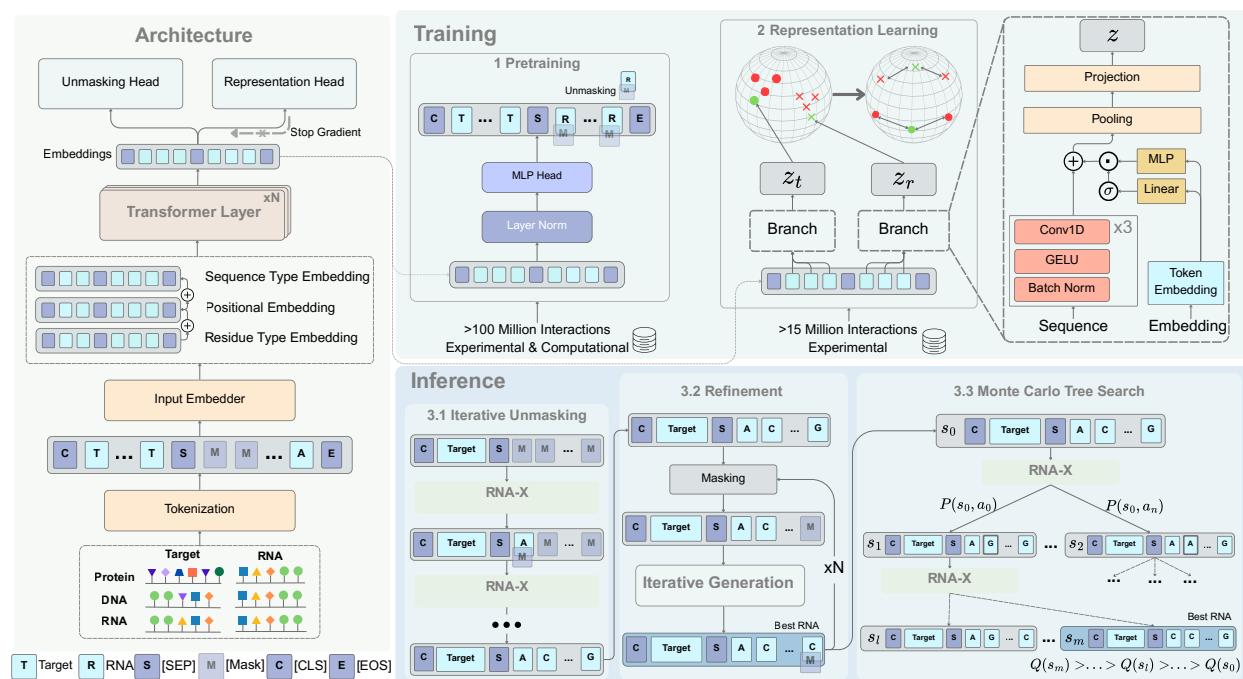


Fig. 1: Overview of RNA-X. **Architecture:** RNA-X encodes paired RNA-target sequences through a shared transformer backbone with sequence-type, positional, and residue-type embeddings. Separate unmasking and representation heads enable both generation and representation learning objectives. **Training:** The model is trained on over 100 million RNA-target interactions. (1) RNA-X is pretrained by unmasking the masked tokens. Then (2) interaction-aware representation learning using contrastive branches for target (z_t) and RNA (z_r) is performed. The representations obtained in pretraining are transformed into another space (transformed globe) to better distinguish interacting pairs and have better downstream performance. Model weights are kept frozen at this stage. **Inference:** During generation, RNA-X takes target sequence(s) (protein, RNA, DNA, or combinations) and fully masked RNA sequence. It designs RNA sequence through iterative unmasking of the tokens from left to right (3.1). The result is repeatedly refined by randomly masking tokens and passing them through iterative unmasking (3.2). Finally, a Monte Carlo Tree Search (MCTS)-based exploration is performed (3.3). The probability P of choosing action a in state s is determined by RNA-X logits, and the value of the state Q is calculated using the scoring function (Supplementary Notes 1.7), which optimizes the designed sequences to bind the target while preserving RNA structural and thermodynamic stability.

Our results show that RNA-X designs natural-like RNAs (i) with matching or better predicted affinity/stability compared to known binders of all target types. It outperforms the state-of-the-art methods in targeting ten proteins. We show that the model can successfully design to target therapeutically relevant molecules such as p53 and thrombin proteins, and for targets without prior interaction data (e.g., SDAD1 protein), enabling the design for virtually any target molecule sequence. For the first time, we designed a novel guide RNA from scratch in a single shot that simultaneously binds to (i) the DNA of *Streptococcus pyogenes*, and (ii) the Cas9 protein. The resulting design achieves a predicted binding energy comparable to that of the wild-type guide RNA reported in the Protein Data Bank and is close to a manual optimization. Despite having orders of magnitude fewer parameters than existing RNA foundation models, our optimized embeddings outperform the state-of-the-art in RNA interaction-related downstream tasks such as estimat-

ing siRNA efficacy and sgRNA on-target knockout efficacy prediction. The code and pretrained model are publicly available on GitHub.

2 Results

2.1 RNA-X designs RNAs with higher predicted binding score compared to other generative models

We compare RNAs designed by RNA-X with RNAs designed by other models when targeting proteins. Please note that there are no other methods in the literature that can be compared when targeting RNA or DNA. For a fair comparison we choose three groups of proteins. First we consider two RNA-binding proteins (RBPs) with substantially different numbers of known RNA interactions from the CLIPdb database: RBM5, which has a relatively small set of 3,917 known interactions, and ELAVL1, which has a much larger dataset of 1,079,145 interactions. This selection allows us to assess how the size of available training data influences the novelty and binding scores of the generated RNA sequences. RNAGen and GenerRNA require fine-tuning for each specific target, which requires substantial computational resources. Consequently, we restrict the benchmarks with these methods to RBM5 and ELAVL1. RNAttranslator and BAnG models do not have this restriction, so we set benchmarks with them on a total of 10 randomly selected proteins. Four proteins are already included in our training data (SRSF1, TARDBP, U2AF2, and AGO2) and the remaining four are entirely excluded from the training set of RNA-X (AATF, DDX21, HNRNPA1 and SDAD1). For the other two models (BAnG and RNAttranslator), the latter set is not explicitly excluded from their training sets, giving them an advantage for better design since the models know these proteins.

RNAttranslator requires only the target sequence, and the BAnG requires the structure of the target, which is predicted using AlphaFold2 when unavailable. We left RNAFlow out of our benchmark as BAnG manuscript clearly demonstrated that they have better performance. We design 100 sequences for each target using each method. Binding scores are found using DeepCLIP [16], which is the most widely used RNA-protein affinity predictor. It was trained in a dedicated training set and tested on an independent test set for each target to ensure unbiased predictions. We also compare the performance of the design against the known natural binders of these proteins and a randomly selected set of 100 natural RNAs as negative controls.

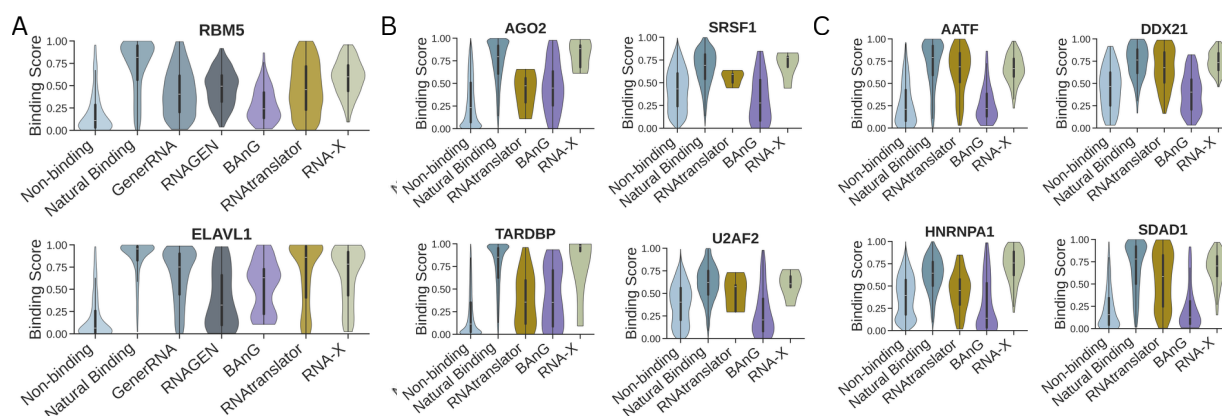


Fig. 2: Comparison of RNA generation models on protein targets (A) Binding score distributions for RBM5 and ELAVL1, two RNA-binding proteins with highly imbalanced numbers of known interactions. **(B)** Benchmark results on four protein targets included in the training data of RNA-X. RNA-X consistently achieves higher predicted binding scores than both BAnG and RNAttranslator. **(C)** Results on four protein targets that were held out entirely from RNA-X's training set but not from BAnG or RNAttranslator's training sets. Despite being evaluated in a true zero-shot setting, RNA-X matches or exceeds RNAttranslator on most proteins and outperforms BAnG in all cases.

Figure 2 shows the results. We observe that the affinity distributions when targeting ELAVL1 are matched for GenerRNA, RNAtTranslator and RNA-X. They are similar to the distribution for known binders of ELAVL1 and are better than RNAGen and BAnG. For RBM5 which have 3 orders of magnitude less training data compared to ELAVL1, RNA-X outperforms all others indicating its ability to generalize in data scarce applications. For the four seen targets, Figure 2B shows that RNA-X consistently outperforms both BAnG and RNAtTranslator. For the remaining six unseen targets, RNA-X achieves stronger performance than RNAtTranslator on three proteins and matches its performance on HNRNPM, while it outperforms BAnG in all cases, as shown in Figure 2C. Note that this is an unfair competition for RNA-X, showing its ability to generalize in zero-shot settings. We further assess whether RNA-X designs RNAs that adopt stable and natural-like secondary structures. Supplementary Figure 5, together with the analysis provided in Supplementary Notes 1.11, show that the designed RNAs show realistic folding behavior, which becomes even more stable and consistent after applying the scoring and exploration stage. We also evaluate the novelty of the designed RNA sequences (See details in Supplementary Note 1.10) and observe that the overall novelty rate for all targets exceeds 90% .

2.2 RNA-X can design binding RNA for Unseen RNA and DNA targets

We further evaluated the binding affinity of the RNA sequences designed by RNA-X on DNA and RNA targets whose interactions were not included during model training. For RNA targets, we selected those with the largest number of known binding RNAs to enable a more statistically robust comparison. In contrast, since each DNA target in the validation set is associated with only a single binding RNA, DNA targets were selected randomly. For each target, we designed the same number of RNA sequences as in the corresponding natural binding RNA set of the target. We used IntaRNA [27] to predict the binding affinity for RNA-RNA interactions, and RIssearch [43] for RNA-DNA interactions. Because no comparable method is available, we compared our designs with the affinities of known natural binder RNAs and with random natural RNAs as negative controls. Please see Supplementary Figure 3A which shows the details of the benchmarking pipelines.

Figure 3 shows the comparative distributions of predicted binding scores for each target type. In Figure 3A, we observe that designed RNAs tend to display strong binding potentials to target RNAs, comparable to or higher than natural binder RNAs. The score distributions are slightly shifted toward lower binding energies for all targets, implying that our designed sequences may adopt optimized structures for interaction. In Figure 3B, we show the results for DNA targets, where the predicted binding energies of designed RNAs are better than natural binder RNAs in two cases and slightly worse in 1 target. In two cases RIssearch did not predict any stable interaction for random or known natural binder RNAs. In those cases the binding score is noted as *NoHit*, reflecting the undetectable binding potential for that sequence. In both cases, our designs achieved negative scores indicating stable binding. We extended this evaluation to a wider set of unseen targets, as shown in Supplementary Figure 3, which includes the same analysis for additional protein, RNA, and DNA targets.

2.3 RNA-X designs RNAs which are predicted to bind to therapeutically important target molecules

We analyze the performance of our model on five therapeutically relevant molecules, including three proteins (p53, thrombin and EGFR), one RNA target (5' UTR of the HIV-1), and one DNA target (3' LTR of the HIV-1). The details of this analysis and the targets are described in Supplementary Note 1.9. For these targets, we compare our designed RNA with two other groups of RNAs: (i) Experimentally validated aptamers which are designed *in vitro* to bind these targets and whose high affinity is confirmed by experimental methods [21,26,44,12,38,35], and (ii) Randomly chosen natural RNAs or RNAs that are experimentally validated to not bind the target [35] as negative controls. We also compare experimentally validated binder RNAs and our designs with respect to their (i) thermodynamic stability, and (ii) secondary structures [24].

Results are shown in Figure 4 (A for P53, B for 5' UTR of the HIV-1, C for HIV-1 3' LTR DNA). Across all targets, the designed RNAs formed stable stem-loop structures with clear paired stems and accessible

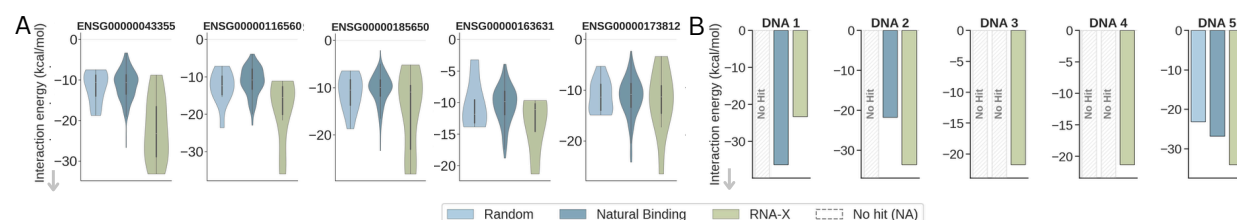


Fig. 3: Evaluation of RNA binding affinity for unseen DNA and RNA targets. Predicted binding score distributions of RNA sequences generated by RNA-X compared to natural binding and non-binding RNAs. **(A)** RNA targets: generated RNAs display strong binding potentials with some targets showing a shift toward lower energies. **(B)** DNA targets: binding affinities of our design predicted by RIssearch are similar to those of known natural binder RNAs; cases where no stable binding was detected are shown as NA.

loops at the predicted interaction sites. These folds are comparable to or more stable than the structures of the validated binder aptamers, as shown by lower normalized MFE values. For p53, the designed RNAs showed stronger MM/GBSA-predicted binding energies [32] and a higher number of hydrogen bonds after molecular dynamics simulation performed in OpenMM [10], compared to validated aptamers and random natural RNAs (Figure 4A). Similar trends were observed for thrombin and EGFR, and their results are shown in Supplementary Figure 4. For the HIV-1 RNA target, our two designs show predicted structures similar to the validated 16nt aptamer and achieved comparable or stronger interaction energies predicted by MM/GBSA and IntaRNA [27]. For the HIV-1 3' LTR DNA targets (sense and anti-sense), the designed RNAs also show stronger binding energies than random baselines and performed similarly to validated DNA-targeting aptamers when evaluated with RIssearch. Notably, in several cases the random RNAs produced no detectable binding signal (No Hit), confirming that they do not form meaningful interactions with the targets.

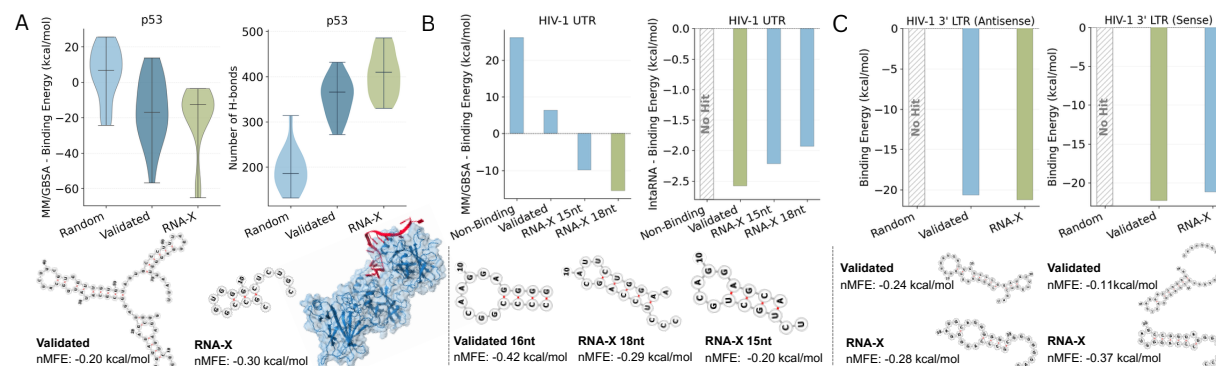


Fig. 4: Evaluation of RNA-X on therapeutically relevant targets. The binding energy distributions of RNA-X on three targets, including a protein target (p53), an RNA target, and a DNA target (HIV-1 3' LTR). Secondary structures of controls and our selected designs and predicted 3D structure of RNA-p53 complex are also shown. **(A)** For the p53 protein, the RNA-X-generated RNAs show lower normalized minimum free energy (nMFE) values, stronger MM/GBSA-predicted binding energies and higher hydrogen bonds during dynamics simulation compared to validated aptamers. **(B)** For the RNA target, RNA-X designs form well-folded and thermodynamically stable secondary structures, with MFE values comparable to or lower than experimentally validated RNA aptamers. These designs also exhibit clear stem-loop architectures characteristic of natural RNA binders. **(C)** For the HIV-1 3' LTR DNA targets (sense and antisense strands), the designed RNAs perform comparably to validated DNA-binding aptamers when evaluated using RIssearch. In all three cases, the designed RNAs show stable folding patterns and energetically favorable interactions.

2.4 RNA-X learns meaningful representations across different interaction types and targets

To evaluate the quality of representations learned by RNA-X, we perform an embedding analysis using balanced RNA-target pairs across three interaction types. We analyze the representations learned by RNA-X using the validation dataset, restricted to experimentally verified interactions to ensure biological reliability. To avoid sampling bias, we construct a balanced subset across the three interaction types, with approximately equal numbers of examples drawn from each. As shown in Supplementary Figure 7A, we visualize the RNA embeddings using t-SNE. The embeddings form well-separated clusters corresponding to different interaction types, indicating that the model captures target-specific information and learns interaction-aware RNA representations.

Within the RNA-protein interaction type, we further visualize the embeddings of RNAs targeting different proteins. In Supplementary Figure 7B, we observe that RNAs interacting with the same protein cluster together, confirming that the learned representations also encode fine-grained target-specific information. In the same figure, we visualize the embeddings of RNAs generated by RNA-X for selected protein targets and compare them with natural RNAs from the dataset. The generated RNA embeddings overlap closely with those of natural RNAs indicating the ability of the model to understand the language of natural RNAs.

We compare both variants of RNA-X with EVO2, RNABERT and AIDO.RNA in RNA-RNA interaction prediction tasks (See Section 4.6 for task details). We use standard RNA interaction datasets such as Huesken, Takayuki, and shRNA as benchmarks. We evaluated all foundation models with frozen backbones. We used identical MLP classifiers with same hyperparameters for fairness during training. As shown in Figure 5, RNA-X achieves the best performance across all datasets, yielding the highest AUROC and AUPRC scores with low variance. Only in shRNA efficiency prediction, AIDO.RNA matches our performance. Note that RNA-X has orders of magnitude smaller number of parameters (135M) compared to EVO2 (1B), and AIDO.RNA (1.6B), yet it is able to learn more informative embeddings. Figure 5 also shows the benefit of optimizing the representations. The performance of RNA-X is consistently lower when its representations are not optimized using the representation head (RNA-X wo-head in Figure 5).

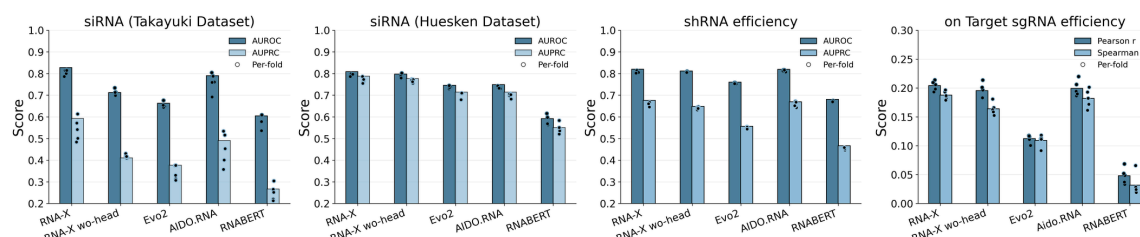


Fig. 5: **Representation learning and downstream performance of RNA-X.** Benchmark results on siRNA, sgRNA and shRNA efficacy prediction tasks. RNA-X achieves the highest AUROC and AUPRC scores compared to existing RNA foundation models. RNA-X wo-head bars show the model's representations without representation head which consistently have lower performances.

2.5 Multi-target RNA Design and Scaffold Optimization

Here, we investigate whether RNA-X can one-shot design RNAs that bind multiple molecular targets simultaneously. We design a novel RNA for the Cas9-DNA-guide RNA complex with PDB ID: 40O8 [28]. The guide RNA was divided into a DNA-complementary spacer and a Cas9-binding scaffold. We generate RNA using RNA-X in three ways to show the flexibility of the model in controlling the generation. We first designed a complete RNA *de novo* by providing both the Cas9 protein and the target DNA as targets (dubbed RNA-X-complete). We use the average predicted binding affinity to both targets as the reward during refinement and MCTS exploration. Second, we designed only the scaffold part *de novo*, while keeping

the spacer part fixed. This tests the model’s ability to selectively redesign structured regions from scratch (dubbed RNA-X-scaffold). Finally, we optimized the scaffold part of the wild type (WT) RNA via refinement and MTCS (not from scratch). We use a composite score that integrates target binding, structural similarity to the wild type RNA, and thermodynamic stability (dubbed RNA-X-scaffold-opt). Full details are provided in the Supplementary Note 1.8.

In the Cas9–DNA–RNA system, the three design modes show different trade-offs relative to the wild type (WT). In MM/GBSA analysis (Figure 6), the RNA-X-complete design shows the least favorable binding energies with both modalities (Figure 6A and B) among others (higher negative energies), indicating weaker stabilization of both the RNA–protein and RNA–DNA interfaces. The RNA-X-scaffold design keeps strong RNA–DNA binding energy as intended (Figure 6B), but has a decrease in RNA–protein affinity (energy) (Figure 6A). The scaffold-optimized sequences (RNA-X-scaffold-opt 1/2/3) clearly improve the balance: They maintain, and in cases outperform, RNA–DNA predicted binding energy of the WT RNA while recovering RNA–protein binding energy. RNA-X-scaffold-opt 3 provides the strongest RNA–DNA binding energy. Figure 6C shows that the RNA-X-complete design has numerous H-bonds with the protein that nevertheless do not translate into favorable MM/GBSA-predicted energy, suggesting suboptimal geometry or contact quality. All designed RNAs have a more negative normalized MFE than WT, indicating higher intrinsic stability; RNA-X-scaffold-opt 2 and 3 show the largest gains (Figure 6E). The views of the Secondary-structure align with these results: the RNA-X scaffold largely preserves the canonical spacer with local scaffold edits, but the RNA-X scaffold-optimized designs form compact stem loops that strengthen Cas9-interacting hairpins without losing spacer complementarity. We also show that RNA-X is more flexible than RNAgen [47] which is a heuristic method that fixes the spacer and optimizes the scaffold with a large pool of point mutations to select the best. On the contrary, RNA-X can optimize or design a completely novel RNA without relying on additional tools. Overall, RNA-X co-optimizes multi-target binding under constraints and yields stable folds while allowing users to choose among conservative (RNA-X-scaffold) or globally optimized (RNA-X-scaffold-opt) designs or completely novel designs (RNA-X-complete). Finally, we also examine the 3D structural differences between the WT and the designed complexes. A summary is provided in Supplementary Figure 8, with details in Supplementary Note 1.8.

3 Discussion

In this work, we introduce RNA-X, the first RNA interaction foundation model and a model for conditional RNA design that learns molecular interactions of RNA with proteins, RNA, and DNA. Unlike previous approaches that focus on a single interaction type, our method learns a shared representation space and a flexible masked-generation strategy to design RNAs for diverse molecular contexts. RNA-X is capable of conditioning on multiple targets simultaneously, demonstrating that the model captures higher-order interaction principles rather than memorizing isolated patterns. Another advantage of our approach is the combination of generative modeling with explicit scoring functions. The iterative mask–predict procedure allows the model to explore local sequence neighborhoods, while the scoring module provides global guidance through binding probability, structural quality, and stability metrics. This combination enables controllable RNA design, which can bias generation toward specific structural features or biochemical constraints simply by adjusting scoring weights. We show that this control is essential when designing functional RNAs such as Cas9 guide RNAs, where both multi-target binding and the preservation of native secondary structure are required for activity. Our results also highlight the importance of representation learning. Without corrective losses, sequence embeddings of different molecules collapse into narrow cones, limiting discrimination among unrelated RNA or protein sequences. By enforcing alignment, whitening, and uniformity, we restore isotropy and produce embeddings that better reflect true biological diversity. These improvements translate into more informative representations, and stronger performance across downstream prediction and generation tasks.

Despite these strengths, several challenges remain. Our study is the lack of experimental validation. Although our computational assessments indicate strong binding and MD simulations provide dynamic insight into complex stability, these methods are inherently imperfect. Molecular dynamics can introduce errors due to initialization choices or boundary effects. Tools such as DeepCLIP offer valuable predictions, but

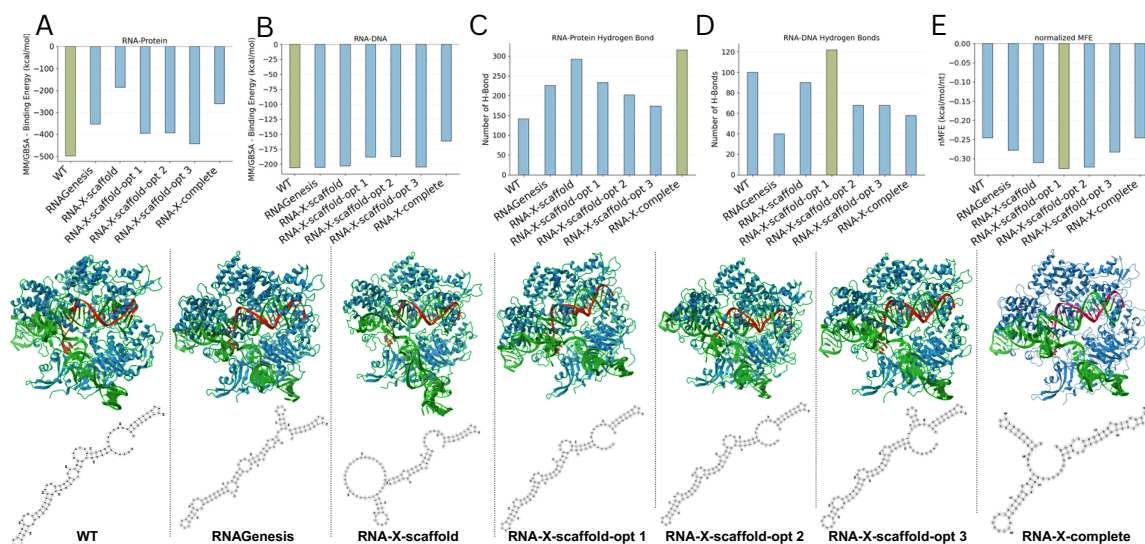


Fig. 6: Multi-target RNA design with RNA-X in the Cas9-DNA-RNA complex. The figure shows the binding energies between RNA-protein (A) and RNA-protein (B) interfaces and the number of hydrogen bonds (C and D) in the Cas9-DNA-RNA complex along with the total normalized MFE of the complex structure (E). The MM/GBSA binding energy analysis indicates that the scaffold-optimized designs (structure-opt-1/2/3) best preserve and enhance dual binding compared to the wild type (WT), while the de novo complete design shows weaker overall stabilization. The hydrogen-bond counts follow the same trend, the spacer-fix design reduces RNA-protein contacts, the structure-optimized designs restore them, and RNA-DNA interactions remain strong across all variants. In addition, all designed RNAs show lower (more negative) normalized MFE values than WT, demonstrating that they adopt more thermodynamically stable structures while maintaining balanced binding to both targets.

they rely on protein-RNA datasets from specific RNA-binding proteins and therefore cannot be generalized to proteins without available training data. Moreover, our framework does not yet explicitly control for designing RNAs that bind at a precise site or motif within the target molecule. Future extensions that integrate site-specific constraints and experimental assays such as surface plasmon resonance or isothermal titration calorimetry could explore for validating and refining the designed interactions.

4 Methods

4.1 Datasets and preprocessing

Data. We curate a dataset of ~ 102 million RNA interactions from databases listed in Supplementary Table 1. The data is filtered and clustered as described in Supplementary Note 1.1. Detailed statistics for interaction types and train/test splits are provided in Supplementary Tables 2, 3 and 4, respectively. An overview of the RNA and protein type distributions in the dataset is shown in Supplementary Figure 2. **Tokenization.** Each amino acid in target proteins and nucleotide in nucleic acids are represented by a single token, yielding a vocabulary of 35 tokens (See details in Supplementary Note 1.2). **Sampling.** Then, to reduce bias due to overrepresented molecules and hubs, we apply a static weighting scheme that penalizes frequent RNAs and targets while increasing the importance of underrepresented molecules. Each interaction is assigned a weight $w = (deg_{RNA})^a (deg_{target})^b (clust_{RNA})^c (clust_{target})^d - \gamma$, where the parameters a, b, c, d and γ regulate the impact of the molecular interaction count (deg) and the size of the cluster (clust). Feature values are normalized by medians in the data set, cut in the 99.5th percentile, and adjusted adaptively based on their dispersion between interaction types. During training, we maintain balance between interaction types by random selection and dynamically controlling the ratio of computational to experimental data. A linear scheduler shifts the focus from computational data early in training to experimental data later for improved

generalization shown in Supplementary Figure 1. Within each subset, samples are drawn with respect to their normalized weights using inverse cumulative sampling (See Supplementary Note 1.3).

4.2 RNA-X Architecture

RNA-X is a transformer-based masked language model designed to jointly model RNA and its molecular targets. It consists of L pre-layer-normalized Transformer encoder layers, each composed of multi-head self-attention and SwiGLU feed-forward sub-blocks. Attention operates over the concatenated RNA-target sequence, enabling bidirectional information exchange across molecule types. Let $r = (r_1, r_2, \dots, r_n)$ be the RNA sequence and $t = (t_1, t_2, \dots, t_m)$ the target sequence. We concatenate the two to form a single input token sequence $s = [t; r]$ of length $n + m$. Each token is represented by the sum of three embeddings—token, position, and residue type—forming the input $e_0^i = \text{Embed}_{\text{token}}(s_i) + \text{Embed}_{\text{pos}}(i) + \text{Embed}_{\text{type}}(s_i)$ where $e_0^i \in \mathbb{R}^d$ is the embedding of token s_i . The embedding type of the residue distinguishes whether a token originates from RNA, protein, or DNA, allowing the model to capture biological context. At the final layer L , a linear projection and softmax produce nucleotide probabilities for each masked site, $p(s_i | \tilde{s}) = \text{softmax}(W e_L^i + b)$ where W is the parameter matrix, b is the bias term and $\tilde{s}$ denotes a corrupted version of the input sequence obtained by replacing a subset of positions with a [MASK] token, the masking procedure is described in the following subsection. Training minimizes cross-entropy loss on masked tokens only, following the standard masked language modeling objective.

4.3 Pretraining by Unmasking

We model the problem as a conditional sequence modeling task, where both the RNA and the target sequence are presented as a single concatenated input to the model. To support both representation learning and conditional generation, we apply masking to both the RNA and the target tokens.

We apply random masking to the combined sequence s by selecting a subset of positions $M \subseteq \{1, \dots, n + m\}$. The model receives the corrupted sequence $\tilde{s}$, in which the tokens at positions M are replaced by a special [MASK] token, and is trained to recover the original tokens at those positions. For all $i \in M$, $\tilde{s}_i = [\text{MASK}]$, and for all $i \notin M$, $\tilde{s}_i = s_i$. To jointly optimize sequence generation and later representation learning (see Section 4.4), we adopt a mixed masking strategy that combines structured and random corruption. For RNA sequences, in 80% of training steps the masking rate is drawn from a $\beta(3, 9)$ distribution (mean $\approx 25\%$), and in the remaining 20%, it is uniformly sampled from $[0, 1]$, resulting in an average mask rate of about 30%. This schedule allows the model to learn from both mild and severe sequence perturbations. For target sequences, a fixed 15% of residues are masked at random, providing a stable conditioning signal during training (See Supplementary Note 1.4 for masking details).

The objective is a single masked language modeling loss: $\mathcal{L} = -\mathbb{E}_{s, M} [\sum_{i \in M} \log p(s_i | \tilde{s})]$. Since the model must predict tokens from both parts of the sequence using shared attention layers, it implicitly learns to model dependencies between RNA and target tokens. This results in embeddings that are sensitive to interaction-specific features rather than isolated sequence patterns.

4.4 Representation Learning

To learn stable and discriminative RNA-target embeddings, RNA-X uses a geometry-aware representation learning scheme. Although dynamic masking during pretraining produces contextual token embeddings, simple pooling often leads to modality-specific collapse, where RNA embeddings cluster with RNA and target embeddings cluster with target, creating anisotropic and low-rank structures. To avoid this, we apply a geometry-corrected projection head to the frozen encoder embeddings $e^{(\text{target})}$ and $e^{(\text{rna})}$, producing compact sequence vectors z_t and z_r via gated fusion and masked mean-max pooling (See the Supplementary Note 1.5 for details).

We train these representations with three complementary losses. (i) We apply a Barlow Twins objective. Let $z_t, z_r \in \mathbb{R}^{N \times d}$ where N is the number of samples and d is the representation size, to be the

batch-standardized target and RNA embeddings, and let $C = (N - 1)^{-1} z_t^\top z_r$ be the inter modality correlation matrix. The loss $\mathcal{L}_{\text{BT}} = \sum_i (C_{ii} - 1)^2 + \lambda_{\text{off}} \sum_{i \neq j} C_{ij}^2$ aligns positive RNA–target pairs while reducing feature redundancy; (ii) To prevent low-rank collapse, we apply the whitening regularization. For any standardized embedding z , we compute $C = (N - 1)^{-1} z^\top z$ which is the intra-modality correlation matrix. We minimize $\mathcal{L}_{\text{white}}(z) = d^{-1} \sum_j (C_{jj} - 1)^2 + d^{-1} \sum_{i \neq j} C_{ij}^2$ and apply it to both modalities: $\mathcal{L}_{\text{white}}^{\text{total}} = \mathcal{L}_{\text{white}}(z_t) + \mathcal{L}_{\text{white}}(z_r)$; and (iii) To distribute embeddings evenly on the hypersphere, we use the uniformity loss $\mathcal{L}_{\text{uni}}(z) = \log \left[\frac{1}{N(N-1)} \sum_{i \neq j} \exp(-2\tau \|z_i / \|z_i\| - z_j / \|z_j\|\|_2^2) \right]$ with $\tau = 2.0$, applied to both modalities. The final objective combines these terms: $\mathcal{L}_{\text{repr}} = w_{\text{bt}} \mathcal{L}_{\text{BT}} + w_{\text{white}} \mathcal{L}_{\text{white}}^{\text{total}} + w_{\text{uni}} \mathcal{L}_{\text{uni}}^{\text{total}}$, which jointly promotes alignment across modalities, isotropy within each modality, and uniform distribution across the representation space, resulting in balanced and interaction-aware embeddings.

4.5 Inference

At inference, RNA–X generates RNA sequences conditioned on a single or multiple target(s) through a three-stage process combining masked generation, refinement, and Monte Carlo Tree Search (See details in Supplementary Note 1.6). The model first performs iterative masked prediction, progressively filling masked regions of an RNA sequence. The generated RNA is then refined through mask–predict loops, where uncertain positions are re-masked and re-predicted until convergence. After each iteration, the candidate RNA is evaluated by a composite scoring function (See Supplementary Note 1.7 for details of scoring), and only the best-scoring RNA is retained for further refinement. This best-first procedure accelerates convergence and stabilizes optimization/ Finally, Monte-Carlo Tree Search (MCTS) is applied for global optimization, where each node represents a candidate RNA sequence and edges correspond to single-base edits. Node selection follows the PUCT rule $U(s, a) = Q(s, a) + c P(s, a) \sqrt{N(s)} / (1 + N(s, a))$, where $Q(s, a)$ is the mean value of action a at state s , $P(s, a)$ is the model-derived prior probability of choosing that action, $N(s)$ is the total number of visits to node s , and $N(s, a)$ is the number of times action a has been taken from s . The constant c controls the balance between the exploration of less-visited edits and the exploitation of high-value moves. Each expansion is guided by the nucleotide probabilities of the model and is immediately evaluated by the same scoring function, guiding the search toward RNA sequences with a predicted higher binding affinity, stability and structural quality.

4.6 Evaluation

Please see Supplementary Note 1.13 for a detailed description of experimental settings. To evaluate the performance of RNA–X, we analyze the binding affinity of the designed RNA sequences and the learned representations. In stability analysis, we analyze sequence-level properties such as GC content, sequence length, and minimum free energy (MFE) and compare their distributions with those observed in natural binder RNAs as baseline and randomly selected natural RNAs as negative controls. To assess binding affinity, we evaluated the binding score by applying predictive models such as DeepCLIP [16] for RNA–protein; IntaRNA[27] for RNA–RNA; and Rsearch [3]for RNA–DNA which estimate the probability of RNA–target interactions [27,43]. For benchmarking conditional RNA generation to target proteins, we compare our method against four recent models: GenerRNA, RNAGEN, BANg and RNAttranslator [30,49,37,25].

In evaluating our learned representations, we use three widely studied benchmark downstream tasks: CRISPR on-target prediction, small interfering RNA (siRNA) efficacy prediction, and short hairpin RNA (shRNA) efficacy prediction. These tasks capture two major types of molecular interactions: RNA–DNA (CRISPR) and RNA–RNA (siRNA and shRNA). We evaluate the performances in classification tasks using AUROC and AUPRC; and the regression tasks using Pearson and Spearman correlations.

The CRISPR on-target prediction (CRI-On) task [8] includes $\sim 15,000$ sgRNAs targeting 1,071 genes in four human cell lines. Each sgRNA guides a Cas protein to a DNA site. The goal is to predict actual editing efficiency. The siRNA and shRNA tasks focus on predicting RNA–RNA interactions involved in gene silencing. The siRNA interaction datasets from Takayuki (702 sequences) [42] and Huesken (2,361 sequences) [22] include binary labels; and the shRNA dataset (2,076 sequences) provides actual efficiency scores [48].

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